

Lee Yeong Kim

Postdoctoral Researcher, University of Vienna

📍 Vienna, Austria ✉ leeyeong.kim@univie.ac.at ☎ 0664 1279250 🔗 leeyeongkim.github.io/my_research
 🆔 0000-0003-3509-4628 🌐 leeyeongkim

Education

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|---------------------------------------|---|-------------------------------------|
| Com-
bined
MS/
PhD | Ulsan National Institute of Science and Technology (UNIST) , Physics
• Advisor: Prof. Bum Suk Zhao. Dissertation: "Study on matter-wave optics under grazing incidence condition with solid surfaces."
• Grade: 4.02/4.3 | Ulsan, Korea
Mar 2016 – Aug 2022 |
| BS | Ulsan National Institute of Science and Technology (UNIST) , Physics (Minor: Energy Engineering)
• Grade: 3.60/4.3
• Full scholarship recipient | Ulsan, Korea
Mar 2011 – Feb 2016 |

Experience

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|---|---|
| University of Vienna , Postdoctoral Researcher
Research advisor: Prof. Markus Arndt
• Studying matter-wave interference of biomolecules in the BIOTIMA project, extending interferometry from peptides (>2,000 Da) towards intact proteins
• Supported by the quantA Postdoctoral Fellowship (2026-2027) | Vienna, Austria
Jan 2026 – present
9 months |
| Career transition period (spouse's postdoctoral stay) , Seeking academic opportunities to continue my research career
• Remained research active: two peer-reviewed publications (2024) and contributions to MOLEC 2024 and ISMB 2025
• Prepared fellowship applications leading to the quantA Postdoctoral Fellowship and an MSCA Seal of Excellence (proposal score 95.8/100) | Zürich, Switzerland
July 2023 – Dec 2025
2 years 6 months |
| UNIST , Postdoctoral Researcher
Research advisor: Prof. Bum Suk Zhao
• Revealed effects of macroscopic surface curvature, grating magnification, and beam emergence in grazing incidence atom scattering, directing two graduate students (Phys. Rev. A 110, 013313)
• Analyzed diffractive mirrors for neutral-atom matter-wave optics (Farad. Disc. 251, 160) | Ulsan, Korea
Sept 2022 – June 2023
11 months |
| Fritz-Haber-Institut der Max-Planck-Gesellschaft , Visiting Student
Research advisors: Dr. Wieland Schöllkopf and Prof. Bum Suk Zhao
• Seven visits: 4 months (2014, 2017), 2 months (2015, 2016), and 10 months (2019).
• Carried out 4He, 4He ₂ , 4He ₃ and D ₂ scattering experiments on the FHI molecular-beam apparatus
• Experimentally tested Babinet's principle in matter-wave diffraction and identified multiple-edge diffraction, a non-destructive mechanism that enhances elastic reflection from microstructured surfaces (PCCP 23, 8030; PCCP 24, 21593 – front cover)
• Observed the transition from quantum reflection to edge diffraction in a periodic array of half planes (Phys. Rev. Lett. 122, 040401 – Editor's Suggestion) | Berlin, Germany
Jan 2014 – Dec 2019
6 years |

UNIST, Undergraduate Research Assistant

Research advisor: Prof. Bum Suk Zhao

- Built Monte-Carlo trajectory simulations of molecules deflected by pulsed non-resonant optical standing waves, showing that rotational-state-dependent polarizability governs the optical dipole force (Phys. Rev. Lett. 115, 223001)

Ulsan, Korea

Mar 2014 – Feb 2016

2 years

Funding and Fellowships

- **2026-Present** – quantA Postdoctoral Fellowship (quantA)
- **2017-2021** – Global Ph.D. Fellowship, The Korean Government (acceptance rate 7.6:1)
- **Jul.-Nov. 2014** – Global Internship Program Scholarship, The Korean Government

Honors and Awards

Seal of Excellence, Marie Skłodowska-Curie Actions Postdoctoral Fellowship Proposal score: 95.8/100	Feb 2026
UNIST Best Research Award Top 5 excellent graduates	Feb 2023
Best Poster Presentation Award	Jan 2021
Best Poster Presentation Award	Apr 2021
Best Oral Presentation Award	Apr 2016
Ulsan Mayor Award Top 5 excellent graduates	Feb 2016

Publications

Comprehensive peak-width analysis in matter-wave diffraction under grazing incidence conditions Lee Yeong Kim , Do Won Kang, Jong Chan Lee, Eunmi Chae, Wieland Schöllkopf, Bum Suk Zhao 10.1103/PhysRevA.110.013313 (Phys. Rev. A 110, 013313)	2024
Diffractive mirrors for neutral-atom matter-wave optics Lee Yeong Kim , Do Won Kang, Sanghwan Park, Seongyeop Lim, Jangwoo Kim, Wieland Schöllkopf, Bum Suk Zhao 10.1039/D3FD00155E (Farad. Disc. 251, 160)	2024
Enhanced elastic scattering of He2 and He3 from solids by multiple-edge diffraction Lee Yeong Kim , Sanghwan Park, Chang Young Lee, Wieland Schöllkopf, Bum Suk Zhao 10.1039/D2CP02641D (Phys. Chem. Chem. Phys. 24, 21593 (Front cover))	2022
Experimental test of Babinet's Principle in matter-wave diffraction Lee Yeong Kim , Ju Hyeon Lee, Yun-Tae Kim, Sanghwan Park, Chang Young Lee, Wieland Schöllkopf, Bum Suk Zhao 10.1039/D0CP05694D (Phys. Chem. Chem. Phys. 23, 8030)	2021
Scattering of adiabatically aligned molecules by nonresonant optical standing waves Lee Yeong Kim , Byung Gwun Jin, Tae Woo Kim, Ju Hyeon Lee, Bum Suk Zhao 10.1126/sciadv.aaz0682 (Sci. Adv. 6, eaaz0682)	2020

- Matter-wave diffraction from a periodic array of half planes** 2019
 Ju Hyeon Lee, **Lee Yeong Kim**, Yun-Tae Kim, Chang Young Lee, Wieland Schöllkopf, Bum Suk Zhao
[10.1103/PhysRevLett.122.040401](https://doi.org/10.1103/PhysRevLett.122.040401) (Phys. Rev. Lett. 122, 040401 (Editor's Suggestion))
- Effect of rotational-state-dependent molecular alignment on the optical dipole force** 2016
Lee Yeong Kim, Ju Hyeon Lee, Hye Ah Kim, Sang Kyu Kwak, Bretislav Friedrich, Bum Suk Zhao
[10.1103/PhysRevA.94.013428](https://doi.org/10.1103/PhysRevA.94.013428) (Phys. Rev. A 94, 013428)
- Rotational-state-dependent dispersion of molecules by pulsed optical standing waves** 2015
 Xing Nan Sun, **Lee Yeong Kim**, Bum Suk Zhao, Doo Soo Chung
[10.1103/PhysRevLett.115.223001](https://doi.org/10.1103/PhysRevLett.115.223001) (Phys. Rev. Lett. 115, 223001)

Co-Authored Publications

- Zhang, Lee, Kim, Jin, Kim, **Kim**, Zhao, and Schöllkopf, "Universal diffraction of atomic and molecular matterwaves: A comparison of He and D2 quantum reflected from a grating," Chem. Phys. Chem. 17, 3670 (2016). DOI: [10.1002/cphc.201600965](https://doi.org/10.1002/cphc.201600965)
- Sun, Jin, **Kim**, Kim, and Zhao, "Strong optical dipole force exerted on molecules having low rotational temperature," Chem. Phys. Chem. 17, 3701 (2016). DOI: [10.1002/cphc.201600838](https://doi.org/10.1002/cphc.201600838)

Technical Skills

- **Laboratory Techniques**
 - Matter-wave scattering experiments of 4He, D2 at 2-100 neV under grazing incidence at high vacuum (10^{-6} to 10^{-12} mbar)
 - Vacuum systems: diffusion, rotary, NEG, and turbo pumps
 - Nanosecond IR lasers (Nd:YAG: Surelite, Powerlite) and visible dye (Sirah) lasers – operation, maintenance, troubleshooting
 - Molecular beams and ion detection techniques: Time-of-Flight mass spectrometry, Velocity Map Ion Imaging, Resonance Enhanced Multiphoton Ionization
 - Atomic Force Microscopy (Dimension ICON, NX10): image acquisition, processing, and analysis
 - Authored operating manuals and standard procedures for daily laboratory experiments
- **Computational Skills**
 - Molecular trajectory calculations with the Monte-Carlo method, solving the Schrödinger equation (C++)
 - Experimental automation using LabVIEW programming
 - Converted and upgraded simulations between C, Python, Mathematica, C++, Matlab, and LabVIEW
 - C++, Matlab, LabVIEW, LaTeX, Origin, Adobe Illustrator

Languages

Korean

Native

English

Fluent; working language of research and publication

German

Intermediate (B1 TELC certificate)

Supervising and Mentoring

- Provided guidance to 4 graduate and 3 undergraduate students in research projects
- Served as teaching assistant (TA) 5 times for experimental physics laboratory courses
- Designed experiments and assisted as TA twice for physical chemistry laboratory courses
- Student representative, Department of Electrical and Electronic Engineering (2012): liaison between students and faculty, organizing departmental meetings and student activities. Later, I changed my major from Device Physics to Physics.

Conference Activities

- **Sep. 2026**, Germany – Oral: Hot topic, 25th European Conference on the Dynamics of Molecular Systems
- **Jun. 2026**, Austria – Poster: Conference on Quantum Fluid Clusters, QFC 2026
- **May 2026**, Austria – Oral: COST COSY WG4 Meeting
- **Jun. 2025**, Switzerland – Poster: 2025 International Symposium on Molecular Beams (ISMB)
- **Jul. 2024**, Poland – Poster: 24th European Conference on the Dynamics of Molecular Systems
- **Oct. 2021**, Korea – Poster: 2021 Korean Physical Society Fall Meeting
- **Jul. 2021**, Online – Oral: Invited talk, 2021 International Symposium on Molecular Beams (ISMB)
- **Apr. 2021**, Korea – Poster: 2021 Korean Physical Society Spring Meeting
- **Mar. 2021**, Online – Oral: The 5th Asian Workshop on Molecular Spectroscopy (AWMS)
- **Nov. 2020**, Korea – Oral: 2020 Society of Global Ph.D. Fellows Annual Conference
- **Nov. 2020**, Korea – Oral: 2020 Korean Physical Society Fall Meeting
- **Nov. 2019**, Korea – Oral: 2019 Society of Global Ph.D. Fellows Annual Conference
- **Oct. 2019**, Korea – Oral, Poster: 2019 Korean Physical Society Fall Meeting
- **Sep. 2018**, Greece – Poster: 2018 Frontiers of Matter Wave Optics (FOMO)
- **Sep. 2017**, Korea – Poster: 2017 Korean-French Joint Symposium
- **Apr. 2017**, Korea – Oral: 2017 Korean Physical Society Spring Meeting
- **Oct. 2016**, Korea – Poster: 2016 Korean Physical Society Fall Meeting
- **Aug. 2016**, Norway – Poster: Scattering of Atoms and Molecules from Surfaces (SAMS)
- **Apr. 2016**, Korea – Oral: 2016 Korean Physical Society Spring Meeting
- **Feb. 2016**, Korea – Oral: The 13th Symposium on Gas-phase Chemical Dynamics & Spectroscopy
- **Aug. 2015**, Korea – Oral: The 12th Symposium on Gas-phase Chemical Dynamics & Spectroscopy
- **Jun. 2015**, Spain – Poster: 2015 Symposium on Molecular Beams (ISMB)
- **Apr. 2015**, Korea – Oral: 2015 Korean Physical Society Spring Meeting
- **Apr. 2015**, Korea – Oral: 2015 Korean Chemical Society Spring Meeting
- **Jun. 2014**, Korea – Poster: The 116th 2014 Korean Physical Chemistry Society Fall Meeting